3	**(a)**	**(i)**		For the car to **stay in contact with the floor, the normal contact force N > 0.** $mg - N = \frac{mv^2}{r}$ $N = mg - \frac{mv^2}{r}$ $N > 0$ $mg > \frac{mv^2}{r}$ $v > \sqrt{gr} > \sqrt{9.81 \times 12.0}$ $v > 10.8 \text{ m s}^{-1}$	B1 M1 A1
		(ii)		By conservation of energy, lost in gravitational p.e. = gain in kinetic energy (energy conservation) $mg\left(2 \square 12\right) = \frac{1}{2}mv^2 - \frac{1}{2}m\left(10.850\right)^2$ $v = 24.3 \text{ m s}^{-1}$	 M1 A1
	(b)	**(i)**		angle turned per unit time *or* rate of change of angle *or* rate of change of angular displacement in a particular direction	 M1 A1
		(ii)		if the car is below minimum speed in **(a)(i)** (or at rest), its **weight** will cause it to fall inwards arm can exert an **upward** force (equal to weight) on the carriage in **(b)** even when it is at rest	B1 B1
					[Total:9]